Engineering Probability
Homework 5

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(1)

Given a random variable X with mean μ_X and variance σ_X^2, find the mean and variance of the standardized random variable

$$Y = \frac{1}{\sigma_X}(X - \mu_X).$$

$$\mu_Y = E[Y] = \frac{1}{\sigma_X}(E[X - \mu_X]) = \frac{1}{\sigma_X}(0) = 0$$

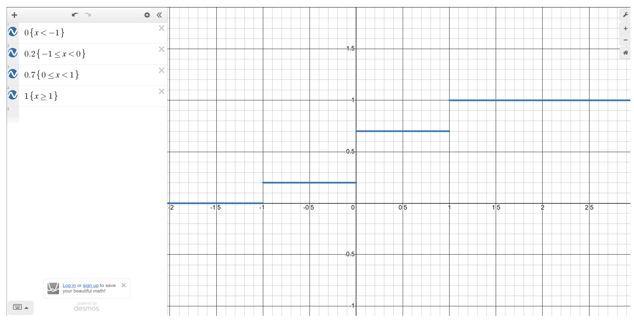
$$\sigma_Y^2 = \mathrm{Var}(Y) = \left(\frac{1}{\sigma_X}\right)^2 (\mathrm{Var(X)}) = \left(\frac{1}{\sigma_X^2}\right)(\sigma_X^2) = 1$$

(2)

The random variable X has CDF

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \le x < 0, \\ 0.7, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

(a) Draw a graph of the CDF.



(b) Write $P_X(x)$, the PMF of X. Be sure to write the value of $P_X(x)$ for all x from $-\infty$ to ∞.

$$P_X(x) = \begin{cases} 0.2, & x = -1, \\ 0.5, & x = 0, \\ 0.3, & x = 1, \\ 0 & \text{otherwise.} \end{cases}$$

(3)

A grocery store sells X hundred kilograms of rice every day, where the distribution of X is:

$$F(x) = \begin{cases} 0, & x < 0, \\ kx^2, & 0 \le x < 3, \\ k(-x^2 + 12x - 3), & 3 \le x < 6, \\ 1, & x \ge 6. \end{cases}$$

Suppose that this grocery store's total sales of rice do not reach 600 kilograms on any given day.

(a) Find the value of k.

$$k(-(6)^2 + 12(6) - 3) = 1$$

$$k(-36 + 72 - 3) = 1$$

$$k(33) = 1 \therefore \boxed{k = \frac{1}{33}}$$

(b) What is the probability that the store sells between 200 and 400 kilograms of rice next Thursday?

$$F_X(4) - F_X(2) = P(2 < X \leq 4)$$

$$P(2 < X \leq 4) = \frac{1}{33}\left(-(4)^2 + 12(4) - 3\right) - \frac{1}{33}(2)^2$$

$$P(2 < X \leq 4) = \frac{1}{33}\left(-(16) + 48 - 3\right) - \frac{4}{33}$$

$$P(2 < X \leq 4) = \frac{1}{33}(29) - \frac{4}{33}$$

$$P(2 < X \leq 4) = \frac{25}{33}\boxed{= 0.75\overline{75}}$$

(c) What is the probability that the store sells over 300 kilograms of rice next Thursday

$$P(X > 3) = 1 - P(X \leq 3) = 1 - F_X(3)$$

$$P(X > 3) = 1 - \frac{1}{33}\left(-(3)^2 + 12(3) - 3\right)$$

$$P(X > 3) = 1 - \frac{1}{33}\left(-(9) + 36 - 3\right)$$

$$P(X > 3) = 1 - \frac{1}{33}(24)$$

$$P(X > 3) = 1 - \frac{1}{33}(24)$$

$$P(X > 3) = \frac{9}{33} = \frac{3}{11}\boxed{= 0.27\overline{27}}$$

(d) Given that the store sold at least 300 kilograms of rice last Friday, what is the probability that it did not sell more than 400 kilograms on that day?

$$P(X \leq 4 | X \geq 3) = \frac{P(X \leq 4 \cap X \geq 3)}{P(X \geq 3)} = \frac{F_X(4) - F_X(3^{-}1)}{1 - F_X(3^{-}1)}$$

$$F_X(4) = \frac{1}{33}(-(4)^2 + 12(4) - 3) = \frac{29}{33}$$

$$F_X(3^{-}) = \frac{1}{33}(3^2) = \frac{9}{33}$$

$$P(X \leq 4 | X \geq 3) = \frac{\frac{29}{33} - \frac{9}{33}}{1 - \frac{9}{33}} = \frac{\frac{20}{33}}{\frac{24}{33}} = \frac{5}{6}$$

(4)

The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0, & x < -1, \\ \dfrac{x+1}{2}, & -1 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Find the PDF $f_X(x)$ of X.

$$f_X(x) = \frac{dF_X(x)}{dx}$$

$$f_X(x) = \begin{cases} 0, & x < -1, \\ \dfrac{1}{2}, & -1 \le x < 1, \\ 0, & x \ge 1. \end{cases}$$

(5)

The probability density function of random variable X is

$$f_X(x) = \begin{cases} \dfrac{1}{2}e^{-x/2}, & x \ge 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) What is $P[1 \le X \le 2]$?

$$\int_1^2 f_X(x)dx = P[1 \le X \le 2]$$

$$P[1 \le X \le 2] = \int_1^2 \left(\frac{1}{2}e^{-\frac{x}{2}}\right) dx$$

$$P[1 \le X \le 2] = \frac{1}{2}\left(-2e^{-\frac{x}{2}}\right)\Big|_1^2$$

$$P[1 \le X \le 2] = \left(-e^{-\frac{x}{2}}\right)\Big|_1^2$$

$$P[1 \le X \le 2] = \left(-e^{-\frac{(2)}{2}}\right) - \left(-e^{-\frac{(1)}{2}}\right)$$

$$P[1 \le X \le 2] = \left(-\frac{1}{e}\right) + \left(\frac{1}{e^{\frac{1}{2}}}\right) \boxed{\approx 0.239}$$

(b) What is $F_X(x)$, the cumulative distribution function of X?

$$F_X(x) = \int_{-\infty}^{x} f_X(x)dx$$

$$F_X(x) = \int_0^x \left(\frac{1}{2}e^{\frac{-x}{2}}\right) dx \text{ (for } x \geq 0)$$

$$F_X(x) = \left(-e^{-\frac{x}{2}}\right)\Big|_0^x = \left(-e^{-\frac{x}{2}}\right) + 1$$

$$\boxed{F_X(x) = \begin{cases} 0, & x < 0 \\ -e^{-x/2} + 1, & x \geq 0. \end{cases}}$$

(c) What is $E[X]$, the expectation of X?

$$E[X] = \int_0^\infty x\frac{1}{2}e^{\frac{-x}{2}}\,dx = \frac{1}{2}\int_0^\infty xe^{\frac{-x}{2}}\,dx$$

D		I
x	$+$	$e^{\frac{-x}{2}}$
1	$-$	$-2e^{\frac{-x}{2}}$
0		$4e^{\frac{-x}{2}}$

$$E[X] = \frac{1}{2}\left(-2xe^{\frac{-x}{2}} - 4e^{\frac{-x}{2}}\right)\Big|_0^\infty$$

$$E[X] = -1\left(xe^{\frac{-x}{2}} - 2e^{\frac{-x}{2}}\right)\Big|_0^\infty$$

$$E[X] = \lim_{t\to\infty}\left(-1\left(te^{\frac{-t}{2}} - 2e^{\frac{-t}{2}}\right)\right) - \left((0)e^{\frac{-(0)}{2}} - 2e^{\frac{-(0)}{2}}\right)$$

$$E[X] = 0 - (-2) = \boxed{2}$$

(d) What is $\text{Var}[X]$, the variance of X?

$$E[X^2] = \int_0^\infty x^2\frac{1}{2}e^{\frac{-x}{2}}\,dx$$

D		I
x^2	$+$	$e^{\frac{-x}{2}}$
$2x$	$-$	$-2e^{\frac{-x}{2}}$
2	$+$	$4e^{\frac{-x}{2}}$
0		$-8e^{\frac{-x}{2}}$

$$E[X^2] = \frac{1}{2}\left(-2x^2e^{\frac{-x}{2}} - 8xe^{\frac{-x}{2}} - 16e^{\frac{-x}{2}}\right)\Big|_0^\infty$$

$$E[X^2] = -1e^{\frac{-x}{2}}\left(x^2 + 4x + 8\right)\Big|_0^\infty$$

$$E[X^2] = \lim_{t\to\infty}\left(-1e^{\frac{-t}{2}}\left(t^2 + 4t - 8\right)\right) - \left(-1e^{\frac{-(0)}{2}}\left((0)^2 + 4(0) + 8\right)\right)$$

$$E[X^2] = (0) - (-1(1)(-8)) = 8$$

$$\boxed{\text{Var}(X) = E[X^2] - E[X]^2 = 8 - 2^2 = 4}$$

(6)

Continuous random variable X has PDF

$$f_X(x) = \begin{cases} \frac{1}{4}, & -1 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Define the random variable Y by $Y = h(X) = X^2$.

(a) Find $E[X]$ and $\text{Var}[X]$.

$$E[X] = \frac{(3 + (-1))}{2} = \boxed{1}$$

$$\text{Var}(X) = \frac{(3 - (-1))^2}{12} = \frac{16}{12} = \boxed{\frac{4}{3}}$$

(b) Find $h(E[X])$ and $E[h(X)]$.

$$h(E[X]) = h(1) = \boxed{\frac{1}{8}}$$

$$E[h(X)] = E[X^2] = \int_{-1}^{3} \frac{1}{4} x^2 dx = \frac{1}{12} x^3 \Big|_{-1}^{3} = \frac{(3)^3}{12} - \frac{(-1)^3}{12} = \frac{28}{12} = \boxed{\frac{7}{3}}$$

(c) Find $E[Y]$ and $\text{Var}[Y]$.

$$E[Y] = E[H(X)] = \boxed{\frac{7}{3}}$$

$$E[X^4] = \int_{-1}^{3} \frac{1}{4} x^4 = \frac{1}{20} x^5 \Big|_{-1}^{3} = \frac{1}{20}(3)^5 - \frac{1}{20}(-1)^5 = \frac{244}{20} = \frac{61}{5}$$

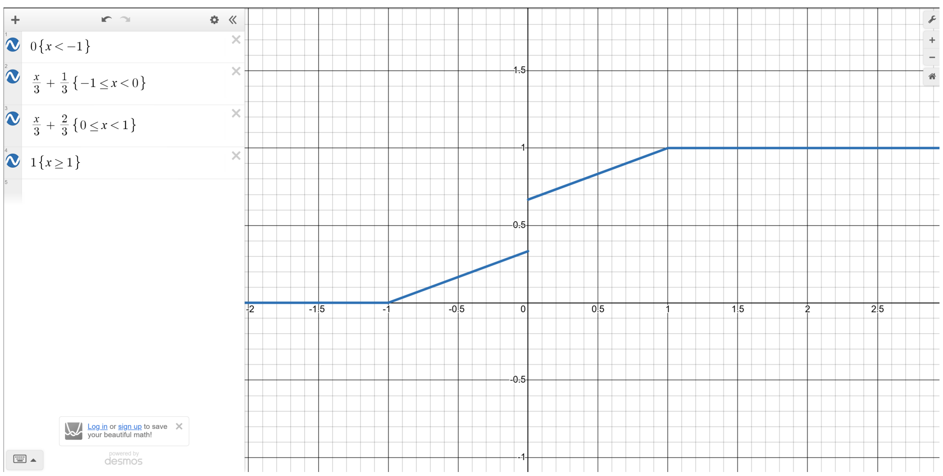
$$\text{Var}[Y] = E[X^4] - (E[X^2])^2 = \frac{61}{5} - \left(\frac{7}{3}\right)^2 = \frac{304}{45} = \boxed{6.7\overline{5}}$$

(7)

Let X be a random variable with CDF

$$F_X(x) = \begin{cases} 0, & x < -1, \\ \dfrac{x}{3} + \dfrac{1}{3}, & -1 \le x < 0, \\ \dfrac{x}{3} + \dfrac{2}{3}, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Sketch the CDF and find:



(a) $P[X < -1]$ and $P[X \le -1]$

$$P[X < -1] = 0 \qquad P[X \le -1] = F(-1) = 0$$

(b) $P[X < 0]$ and $P[X \le 0]$

$$P[X < 0] = F(0^-) = \frac{1}{3}, \qquad P[X \le 0] = F(0^+) = \frac{2}{3}.$$

(c) $P[0 < X \le 1]$ and $P[0 \le X \le 1]$

$$P(0 < X \le 1) = F(1) - F(0^+) = 1 - \frac{2}{3} = \frac{1}{3}, \qquad P(0 \le X \le 1) = F(1) - F(0^-) = 1 - \frac{1}{3} = \frac{2}{3}.$$

(d) $f_X(x)$

$$f_X(x) = \frac{dF_X}{dx} = \frac{1}{3}\delta(x) + \begin{cases} 0, & x < -1, \\ \frac{1}{3}, & -1 \le x < 0, \\ \frac{1}{3}, & 0 < x < 1, \\ 0, & x \ge 1, \end{cases}$$

(e) $E[X]$

$$E[x] = 0$$

(PDF is symmetric about x=0)

(f) $\mathrm{Var}[X]$

$$\begin{aligned}
E[X^2] &= \int_{-1}^{0} \frac{1}{3}x^2\,dx \;+\; 0^2 \cdot P(X=0) \;+\; \int_{0}^{1} \frac{1}{3}x^2\,dx \\
&= \frac{1}{3}\left(\frac{x^3}{3}\Big|_{-1}^{0}\right) + \frac{1}{3}\left(\frac{x^3}{3}\Big|_{0}^{1}\right) \\
&= \frac{1}{3}\left(\frac{1}{3}\right) + \frac{1}{3}\left(\frac{1}{3}\right) = \frac{2}{9} \\
\mathrm{Var}(X) &= E[X^2] - \big(E[X]\big)^2 = \frac{2}{9} - 0^2 = \boxed{\frac{2}{9}}
\end{aligned}$$